

Summary Data on Acid Cropland and Lime Requirements in Zambia

GAIA

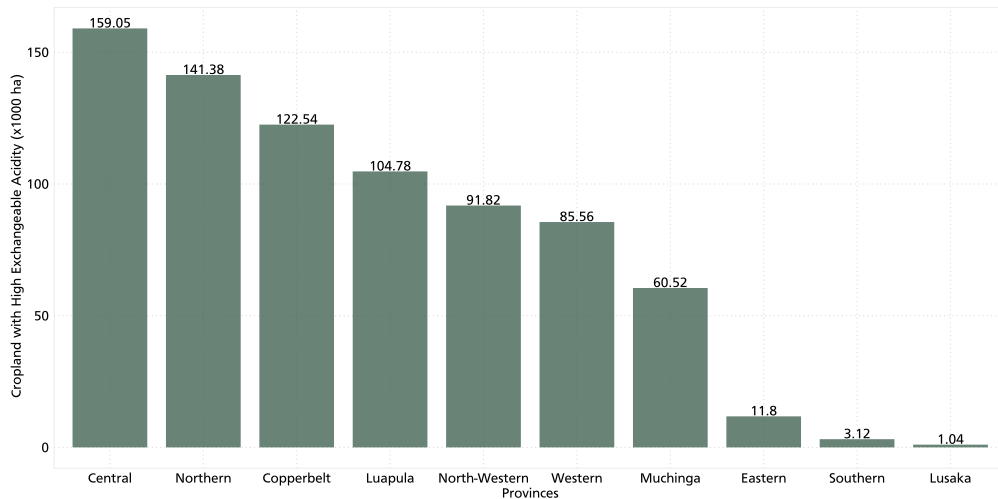
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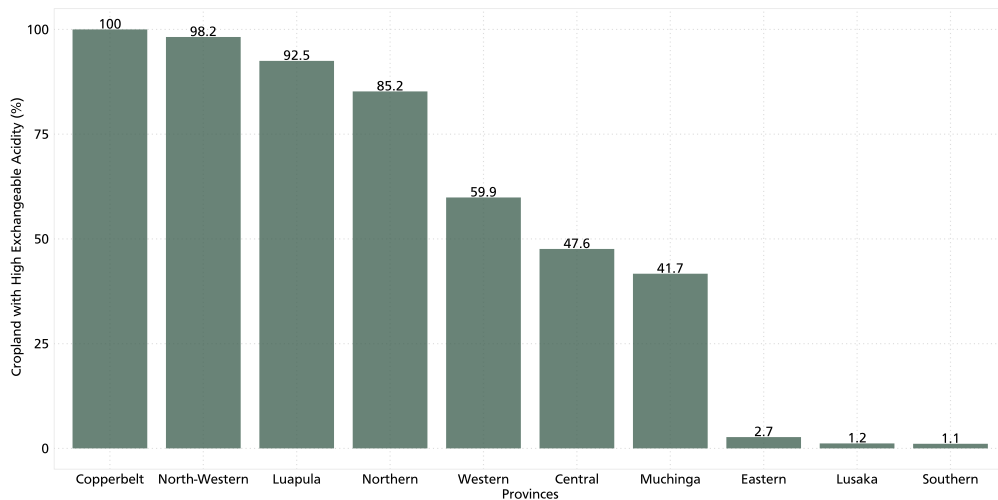
Outlook

- 1 Introduction
- 2 Acidic Cropland
- 3 Yield Loss
- 4 Lime Requirement
- 5 Yield Response
- 6 Additional Production
- 7 ROI for Lime Application
- 8 Profitability of Lime Application

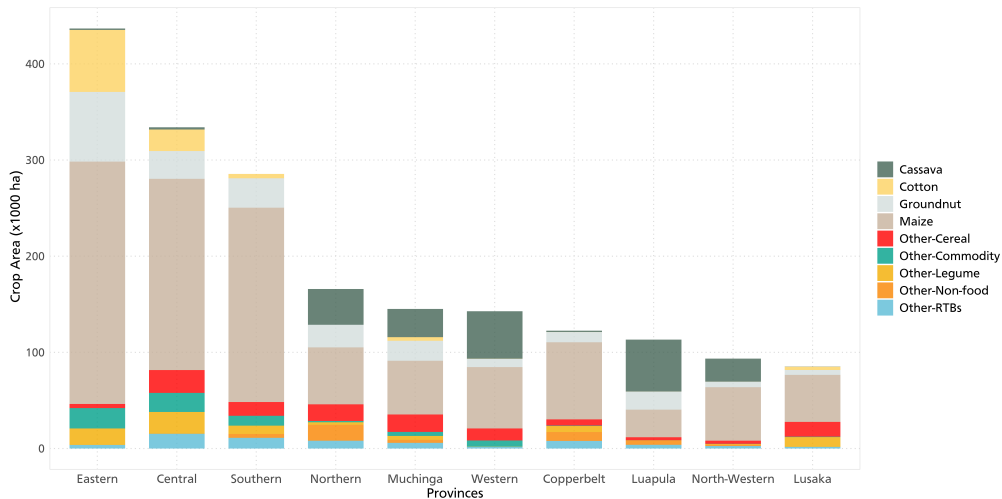
Top 10 Provinces in Zambia with the highest acidic cropland



Top 10 Provinces in Zambia with the highest acidic cropland (Percentage of total cropland)



Total crop area in the top-10 most acidic Provinces, not segregated by acidic cropland



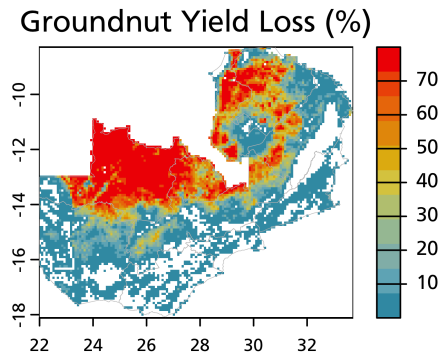
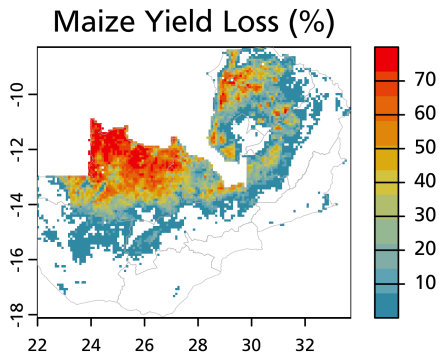
Zambia-Summary

└ Total crop area distribution (ha)

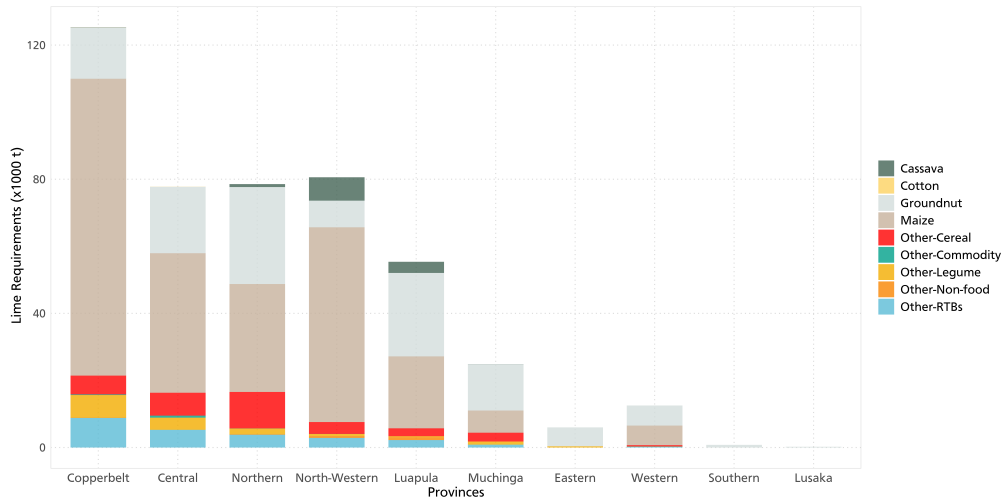
└ Total crop area in the top-10 most acidic Provinces, not segregated by acidic cropland

Provinces	Cassava	Cotton	Groundnut	Maize	Other-Cereal	Other-Commodity	Other-Legume	Other-Non-food	Other-RTBs
Central	2.34	22.37	28.90	198.87	23.77	19.84	22.56	0.10	15.34
Copperbelt	1.34	0.08	10.61	80.24	6.38	0.50	6.30	9.21	7.88
Eastern	1.53	64.49	72.46	252.01	4.28	21.19	17.10	0.03	3.71
Luapula	53.98	0.03	18.78	28.68	3.16	0.06	0.38	4.31	3.85
Lusaka	0.23	3.32	5.11	48.92	15.32	0.42	10.20	-	1.72
Muchinga	29.36	3.75	20.84	55.85	18.12	4.15	4.12	3.13	5.86
North-Western	23.98	0.02	5.71	55.65	3.24	0.14	0.56	1.57	2.61
Northern	37.24	0.01	23.50	59.24	17.29	1.32	2.67	16.52	8.12
Southern	0.05	4.47	30.62	202.04	14.35	10.26	8.69	3.97	11.09
Western	49.36	0.09	8.66	63.77	12.52	6.35	0.18	-	1.81

Yield loss due to soil acidity for Maize and Groundnut in Zambia



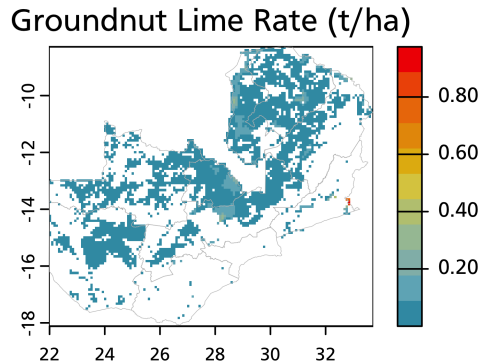
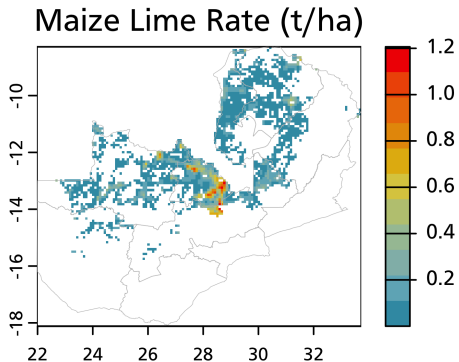
Lime requirements (t) for acid soil remediation in Zambia



Lime requirements (t) for acid soil remediation in Zambia (table)

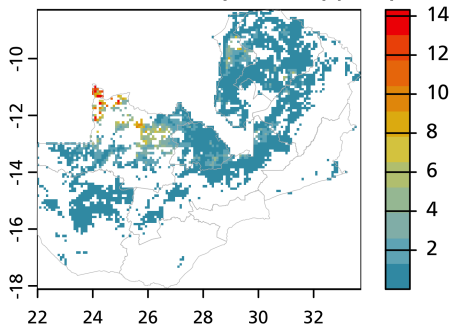
Provinces	Cassava	Cotton	Groundnut	Maize	Other-Cereal	Other-Commodity	Other-Legume	Other-RTBs	Other-Non-food
Central	0.02	0.03	19.79	41.61	6.83	0.60	3.63	5.27	-
Copperbelt	0.11	0.02	15.16	88.52	5.60	0.18	6.76	8.83	0.07
Eastern	-	-	5.67	-	-	-	0.35	-	-
Luapula	3.33	-	24.87	21.46	2.35	0.01	0.32	2.22	0.82
Lusaka	-	-	0.23	-	-	-	-	-	-
Muchinga	0.05	-	13.77	6.62	2.61	-	0.94	0.85	-
North-Western	6.98	-	7.95	58.07	3.53	0.03	0.62	2.86	0.54
Northern	0.90	-	28.92	32.16	10.88	0.02	1.73	3.81	0.12
Southern	-	-	0.78	-	-	-	-	-	-
Western	-	-	5.99	5.83	0.38	0.07	0.03	0.22	-

Lime rate for Maize and Groundnut in Zambia

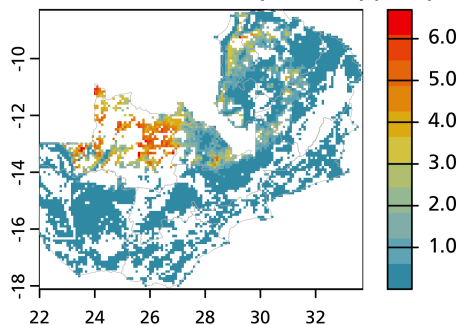


Yield response to lime application in Zambia

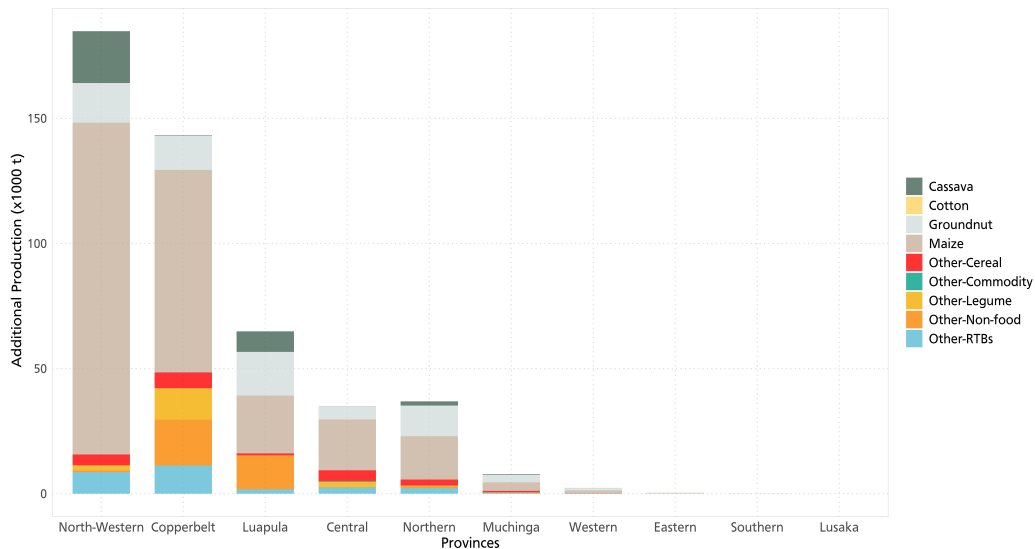
Maize Yield Response (t/ha)



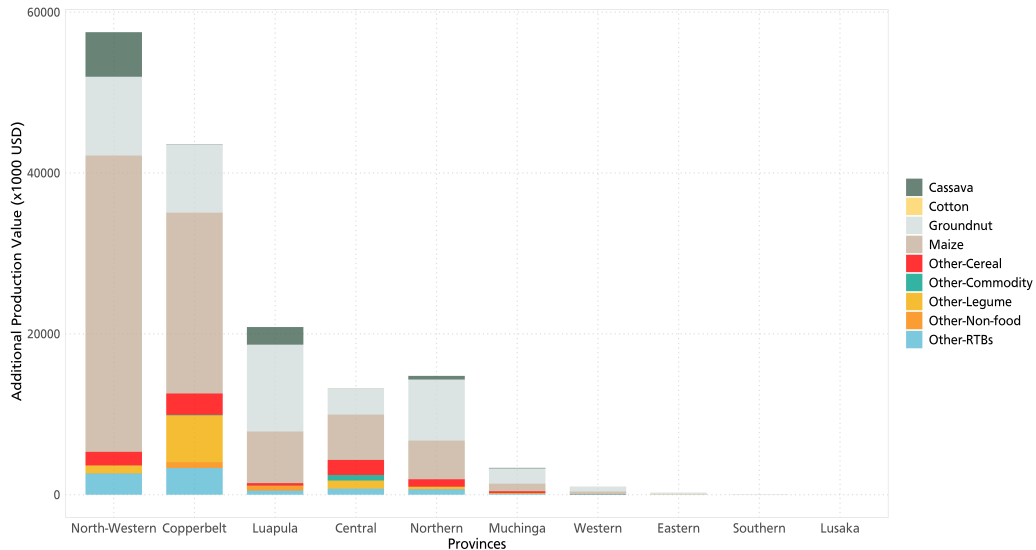
Groundnut Yield Response (t/ha)



Potential for additional production (tone) in top 10 acidic Provinces in Zambia

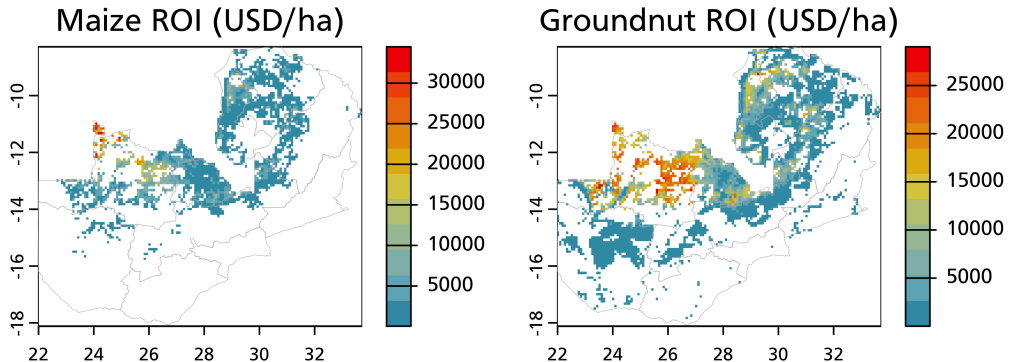


Potential for additional production value (US\$) in top 10 acidic Provinces in Zambia



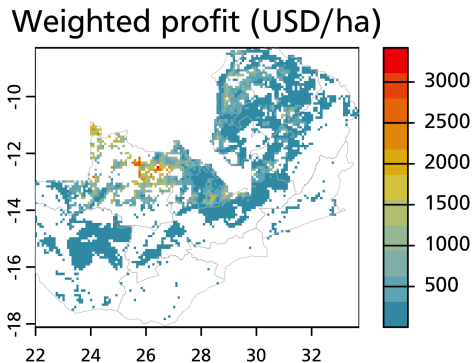
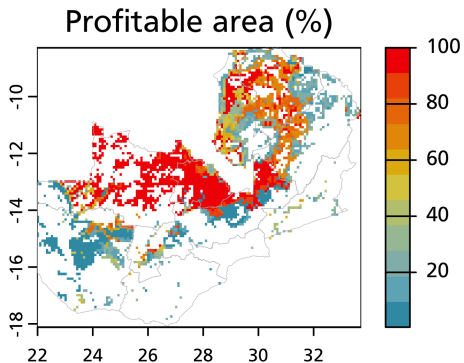
- ROI for Lime Application
- Return on investment for lime application in Zambia

Return on investment for lime application in Zambia



- └ Profitability of Lime Application
- └ Profitability of lime application in Zambia

Profitability of lime application in Zambia



Main assumptions and parameters

- Acidic cropland is defined as having acidity saturation levels above 10% of the Effective Cation Exchange Capacity (ECEC).
- Crop yields are unaffected by soil acidity until a specific acidity saturation threshold Target Acidity Saturation(TAS) is reached; beyond this threshold, yields decrease linearly, eventually reaching zero at high acidity levels.
- The TAS that separates maximum yield from yield decline is used as a critical value for making liming recommendations.
- Actual yields in acid tropical soils are a fraction of the potential yields in non-acidic soils, determined by the estimated yield loss due to acidity.
- Economic benefits from liming are calculated by multiplying the additional yield after soil remediation by median crop prices from sub-Saharan Africa (2016-2020 FAOSTAT data). The cost of liming is calculated using estimated lime application rates and a fixed lime price of 100 US\$/ton.
- The benefits of liming are assessed only for the year of application due to a lack of robust, large-scale data on long-term effects in sub-Saharan Africa.

For details, visit www.acidsoils.africa.